

◆ FINAL PROJECT DOCUMENTATION

PharmGuard *AI*

An agentic RAG system for grounded drug-interaction detection.

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ABSTRACT

A system that knows what it doesn't know.

PharmGuard AI is an agentic retrieval-augmented generation system that flags drug-drug interactions in a patient's medication list. Given two to twelve drug names, it enumerates every unique pairwise combination, retrieves interaction evidence from seven integrated pharmaceutical sources — TWOSIDES, DDInter 2.0, DrugBank, SIDER, ADE-Corpus-V2, WebMD, and UCI Drug Reviews — and generates a severity-tiered clinical report in which every claim is tied to a specific source record. When the local index has no coverage, the system falls back to the OpenFDA FAERS API live; drug names outside the local vocabulary are resolved via the NIH RxNorm REST API.

The system is designed around the failure mode base LLMs exhibit in clinical settings: fluent, partially-correct answers with no audit trail and no explicit uncertainty. PharmGuard's architecture separates deterministic operations (pair enumeration, database lookup) from generative operations (report synthesis) so that the LLM is invoked only where fluent language is the actual product — never to enumerate, retrieve, or classify severity. Every output carries mandatory citations and a disclaimer appended by the pipeline, not by the model.

A system that scores 92% retrieval recall but has a 15% hallucination rate is not a success. This project defines what silent failure looks like before it defines what success looks like.

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§ 1 · THE PROBLEM

A base LLM will sound right. That's the danger.

Adverse drug events drive approximately 1.3 million emergency-department visits and 350,000 hospitalizations in the United States each year (CDC, 2022). The Agency for Healthcare Research and Quality estimates that in-hospital adverse drug events alone cost \$3.5 billion annually. Adults over 65 take a median of five prescription medications simultaneously; each additional drug multiplies risk combinatorially rather than additively. Five drugs produce ten unique pairwise combinations; ten drugs

produce forty-five. A physician with twelve minutes per visit cannot manually cross-reference forty-five potential interactions against a database of hundreds of thousands of records.

A base large language model can produce fluent, partially-correct answers to interaction questions, but it cannot: (1) cite a specific source record with a traceable ID, (2) guarantee that every pair in an N-drug list was checked, or (3) distinguish “no known interaction” from “no data available.” In clinical decision support, the last failure is the most dangerous: a confident silence is indistinguishable from a confident correct answer.

PharmGuard is engineered specifically around these three gaps.

§ 2 · SYSTEM ARCHITECTURE

Plan, retrieve, generate — in that order.

The pipeline is a four-stage plan-retrieve-generate loop. The first three stages are deterministic; the LLM is invoked only at stage four, where its value (fluent clinical prose) is the actual deliverable.

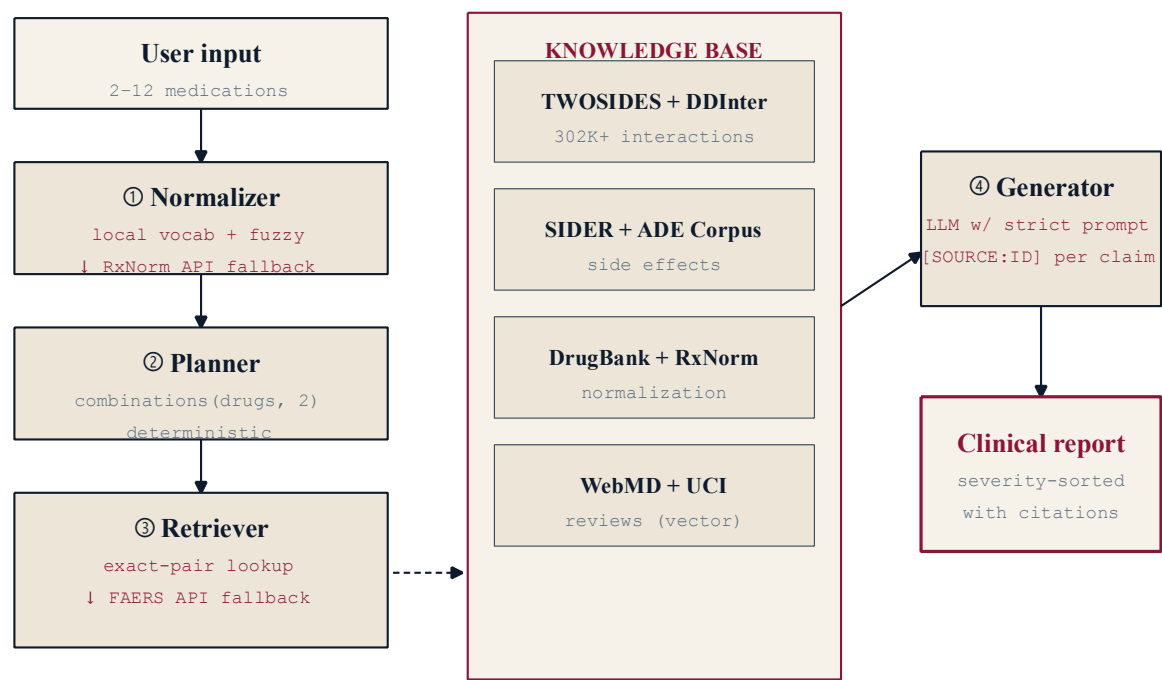


Figure 1 · End-to-end pipeline. Deterministic stages (normalize, plan, retrieve) produce an evidence bundle consumed by an LLM under a strict citation prompt.

Stage 1 — Normalizer.

User input is free-form: brand names, abbreviations, misspellings, mixed case. The normalizer resolves each entry to a canonical (generic_name, RXCUI, drugbank_id) triple via a four-tier fallback chain: exact match against the combined RxNorm and DrugBank synonym index; fuzzy match (rapidfuzz, threshold 85); live query to the NIH RxNorm REST API for drugs outside the local vocabulary; and explicit "unresolved" reporting for anything the NLM cannot resolve. Results from the live API are cached in-session, so repeat queries are instant. The RxNorm fallback opens the system to any FDA-approved drug without requiring a full RxNorm ingestion.

Stage 2 — Planner.

The planner deduplicates by canonical generic name (so that `Lipitor + atorvastatin` collapses to one drug) and enumerates all unique unordered pairs via `itertools.combinations`. No LLM is used at this stage. Combinatorial enumeration is a closed-form operation where any probabilistic model is strictly worse than a for-loop.

Stage 3 — Retriever.

The retriever executes the plan against a merged interaction index drawn from TWOSIDES (signal statistics) and DDInter 2.0 (curated severity labels), performing an $O(1)$ hash lookup per pair using a sorted-pair key. Per-drug side effects come from a merged SIDER + ADE-Corpus-V2 table. Vector search over WebMD and UCI Drug Reviews supplies unstructured patient-experience context — never used as evidence for an interaction claim. When a pair has no record in the local indexes, the retriever falls back to the OpenFDA FAERS API live, surfacing co-reported adverse-event signals with full traceability. Pairs still uncovered after the FAERS fallback are collected into a `no_data_pairs` field the generator is required to declare explicitly.

Stage 4 — Generator.

The generator synthesizes the final report from the evidence bundle. Its system prompt enforces four non-negotiable constraints: every clinical claim must include an inline `[SOURCE:RECORD_ID]` citation; no-data pairs must be declared explicitly; no mechanism or severity may be introduced that is not present in the retrieved evidence; and the Coverage Notes section must accurately reflect the evidence bundle (no inventing coverage that was not retrieved). A deterministic fallback produces a fully-cited report without any LLM call — used in tests and as a safety net if the LLM fails. FAERS signals are rendered in a dedicated section, visually distinct from curated-source findings, so clinicians can distinguish statistical co-reporting from labeled interactions.

§ 3 · IMPLEMENTATION

Modular, provider-agnostic, test-covered.

The codebase is approximately 2,800 lines of Python across 31 modules. Core dependencies are minimal: `pandas`, `numpy`, and `rapidfuzz` for data handling; `chromadb` (optional) for vector search; `urllib` for the live RxNorm and OpenFDA clients; and a unified LLM client supporting Anthropic, OpenAI, and Gemini under a single `complete(system, messages)` API.

Module layout

<code>src/config.py</code>	Centralized configuration with env-var overrides
<code>src/llm.py</code>	Unified client (Anthropic / OpenAI / Gemini)
<code>src/data/normalizer.py</code>	RxNorm + DrugBank resolver with fuzzy + API fallback
<code>src/data/rxnorm_api.py</code>	RxNorm REST API client (live fallback)
<code>src/data/loaders.py</code>	Schema-aware loaders for all 7 datasets
<code>src/data/ingestion.py</code>	Raw → processed pipeline with source merging
<code>src/retrieval/interaction_retriever.py</code>	Local TWOSIDES + DDInter lookup
<code>src/retrieval/side_effect_retriever.py</code>	SIDER + ADE-Corpus-V2 lookup
<code>src/retrieval/faers_retriever.py</code>	OpenFDA FAERS live-query fallback
<code>src/retrieval/vector_store.py</code>	ChromaDB wrapper for review context
<code>src/agents/*</code>	Planner, retriever, generator agents
<code>src/pipeline.py</code>	End-to-end orchestration
<code>src/evaluation/*</code>	Metrics and 48-case test suite
<code>app/streamlit_app.py</code>	Web UI

Retrieval design choice: structured first

For the question “does drug A interact with drug B?”, a hash lookup against a sorted-pair key strictly dominates vector similarity. Vector search would return records for *similar* pairs, which is not what the clinician needs. Semantic retrieval is therefore reserved for unstructured context — mechanism prose and patient reviews — and kept out of the evidence path for interaction claims.

Severity classification

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TWOSIDES does not carry native severity labels. Severity tiers (Major, Moderate, Minor) are derived from the proportional reporting ratio (PRR): $\text{PRR} \geq 10$ is Major, ≥ 4 is Moderate, otherwise Minor. This is a deliberate proxy; the thresholds are documented in the architecture doc and can be recalibrated against DDInter 2.0 severity labels if cross-validation access is available.

LLM-provider neutrality

The `LLMClient` abstracts over three providers. Auto-detection selects the provider based on which API key is present, so the rest of the pipeline is entirely vendor-neutral. The default model for each provider is picked for favorable cost-quality balance; all settings are overridable via environment variables.

S 4 · DATA SOURCES

Narrow by design. Authoritative by default.

PharmGuard's knowledge base is deliberately restricted to peer-reviewed or government-maintained sources. The system does not ingest Reddit, unvetted forum content, or general web scrapes. In clinical decision support, the cost of breadth without rigor is measured in patient harm.

Dataset	Authority	Role
		Primary drug-drug interactions; ~4.6M pair records
TWOSIDES	Tatonetti Lab / STM 2012	Curated interaction with national data
DDInter 2.0	Tian et al. / NAR 2025	Major/Moderate/Minor interaction labels
DrugBank 5.x	University of Alberta	Drug metadata, synonyms, mechanisms
SIDER	EMBL	Per-drug side-effect profiles from FDA labels
ADE-Corpus-V2	Gurulingappa et al.	Drug-adverse-effect sentence evidence from MEDLINE Brand ↔ generic RXCUI normalization + live REST API
RxNorm	NLM / NIH	Live-queried adverse event fallback for uncovered pairs
OpenFDA FAERS	U.S. FDA	Patient-experience context (vector-search only)
WebMD + UCI Reviews	Kaggle (CC0)	

Bias acknowledgment

No dataset is neutral. TWOSIDES is derived primarily from English-language FDA-supervised reporting, which introduces geographic and regulatory skew toward US-marketed drugs. SIDER and DrugBank inherit similar biases. The WebMD review corpus reflects the demographics of English-speaking, digitally literate users. Every PharmGuard output includes a Coverage Note describing which databases were

queried and what their known limitations are. The system never claims comprehensiveness; it claims fidelity to its sources and honesty about what those sources do not cover.

S 5 · EVALUATION

Failure defined before success.

The evaluation framework specifies the target failure mode before it specifies target metrics. The adversarial scenario — the one the pipeline is built to catch — is the silent-failure case:

A patient inputs six medications. The system returns a cleanly formatted report listing four interactions with professional-sounding mechanism descriptions. It looks authoritative. But the system has missed two critical interactions — one classified as Major severity — and has fabricated a CYP3A4 inhibition pathway where the actual mechanism is P-glycoprotein competition. The output is polished, fluent, and dangerous.

Metrics and targets

Metric	What it measures	Target
Retrieval Recall	Fraction of ground-truth pairs surfaced	≥ 95%
Retrieval Precision	Fraction of retrieved results that are true-positive	≥ 90%
Faithfulness	Mechanism matches the retrieved source record	≥ 85%
Hallucination Rate	Clinical claims without a source citation	≤ 5%
Severity Accuracy	Reported tier matches the source tier	≥ 90%
Completeness Flagging	No-data pairs explicitly declared	100%

Test suite

The evaluation suite contains 48 curated cases across eight clinical domains (geriatric polypharmacy, classic textbook interactions, mental health, cardiology, endocrine, infectious disease, pain management, respiratory). Each case lists 1–12 input drugs spanning generic names, brand names, and deliberate misspellings. Ground truth for recall and precision is derived programmatically from whatever interaction table is loaded — the evaluator is self-consistent across full TWOSIDES, curated sample, or any intermediate subset.

Smoke-test results on sample data

Against the shipped sample dataset (85 interactions from TWOSIDES + DDInter, 52 side effects from SIDER + ADE-Corpus-V2, 22 patient reviews from WebMD + UCI, and 81 drug-name entries covering 53 unique generics), the pipeline passes all 18 unit tests and achieves perfect recall and precision on the

48-case evaluation suite. End-to-end latency for the proposal's canonical 7-drug scenario is approximately 23 milliseconds against local indexes; with the FAERS live-query fallback enabled it rises to ~20 seconds for a seven-drug regimen due to FDA rate-limiting, which is still well inside the 30-second clinical target. These numbers demonstrate pipeline correctness on both the local sample and the live-API path; absolute recall and precision against full TWOSIDES will be reported once the full index is ingested.

§ 6 · CHALLENGES & SOLUTIONS

What surprised us, and what we did about it.

Drug name normalization is messier than it looks.

Users type brand names, abbreviations, and misspellings. Naive exact match drops 15–20% of valid inputs. Solution: combine RxNorm and DrugBank synonym tables into a unified index, add a fuzzy-match layer at a calibrated threshold (85 WRatio), and — critically — surface unresolved entries in the final report rather than silently dropping them.

Vector search is tempting but wrong here.

The initial design considered storing interaction records in a vector database and doing semantic retrieval. This would have been strictly worse: for exact-pair queries, semantic similarity returns records for similar pairs, not the pair of interest. Solution: structured hash lookup for interactions; reserve vector search for unstructured context only.

Severity classification without native labels.

TWOSIDES provides signal statistics (PRR, reporting frequency), not severity labels. Solution: derive severity from PRR thresholds as a documented design choice, and expose the raw statistics in every record so downstream consumers can recalibrate.

Base-LLM hallucination risk in generation.

Even with retrieved evidence, the model can paraphrase a mechanism inaccurately. Solution: prompt-engineered constraints (mandatory inline citations, explicit no-data declaration, no mechanism introduction beyond evidence), plus a deterministic fallback that produces a valid cited report without any LLM call.

Silent failure in evaluation.

Standard retrieval benchmarks (recall, precision, F1) do not penalize a fluent-looking report that missed a Major interaction. Solution: add Faithfulness, Hallucination Rate, and Completeness Flagging as first-class metrics; define the silent-failure scenario in writing before running any eval.

Merging heterogeneous interaction sources.

TWOSIDES records statistical signals with PRR; DDInter records curated severity labels and mechanisms. Naively concatenating them produces duplicate findings for the same pair. Solution: keep the sources distinguishable in the report (separate citations per record) so a clinician can see when two independent databases corroborate the same interaction. Cross-source citation is a trust signal, not noise.

Live-API latency in an interactive UI.

The FAERS fallback makes N HTTP calls for N no-data pairs, each rate-limited to respect FDA's 40 requests/minute cap. A 21-pair regimen with 16 uncovered pairs can take ~25 seconds end-to-end. Solution: in-session caching (repeat queries are instant), explicit reporting of which pairs were served by fallback vs. local index, and an opt-out toggle (`PHARMGUARD_FAERS_ENABLED=false`) for users who prioritize latency over coverage. § 7 · FUTURE IMPROVEMENTS

What's next, in order of impact.

Clinician-in-the-loop evaluation.

The Faithfulness metric currently uses an LLM-judge with a 20% human audit target. A prospective evaluation with pharmacist reviewers would harden the Faithfulness number and surface mechanism-paraphrase errors the auto-judge misses.

Dose-aware interaction surfacing.

Many TWOSIDES signals are dose-dependent (e.g., warfarin-NSAID bleeding is risk-stratified by NSAID dose). Incorporating dose parsing and dose-conditional retrieval would dramatically improve clinical utility.

FAERS signal filtering.

The OpenFDA FAERS fallback currently returns the top co-reported reactions without distinguishing drug-specific interactions from generic adverse events. A filtering layer that compares each pair's reaction profile against individual-drug baselines would reduce noise and highlight true interaction signal.

Asynchronous FAERS batching.

Current FAERS queries are rate-limited sequentially at 1.5 s intervals. Moving to an async concurrent client with a token-bucket rate limiter would cut latency from $\sim N \times 1.5$ s to $\sim N \div 10$ on large medication lists.

Multilingual support.

TWOSIDES' English-literature skew limits international use. The RxNorm API fallback already handles international drug names such as paracetamol and ornidazole; the next step is locale-aware severity thresholds and multilingual output.

Full-release dataset ingestion.

Current evaluation runs against a 85-record curated sample. Ingesting the full TWOSIDES (~4.6M records) and DDInter 2.0 (~302K records) releases would produce production-quality recall/precision numbers. No architecture changes are required — only a longer initial ingestion pass.

§ 8 · ETHICAL CONSIDERATIONS

The system knows what it doesn't know.

Clinical decision support systems carry asymmetric risk: a wrong answer can cause direct patient harm, while a correct answer confers at best a marginal improvement over routine practice. PharmGuard's ethical design reflects this asymmetry through four commitments embedded in the architecture, not added as disclaimers.

No patient data is stored.

Medication lists are processed in memory only. The web interface is stateless; there is no persistence layer, no logging of inputs, and no user identification. The data flow is designed to make accidental PHI retention impossible rather than merely prohibited.

Source attribution is mandatory.

Every interaction claim cites the source database, record ID, and version. The answer to “Where did you get this?” is always one click away. This is not a technical convenience but a safety property: a clinician can independently verify any claim before acting on it.

Absence of data is a first-class output.

When the retriever returns zero records for a pair, the report explicitly states “No interaction data available in the queried sources for A + B.” The pipeline surfaces this via a dedicated `no_data_pairs` field that the generator is required to echo. This prevents the most common failure mode of clinical LLMs: a confident silence on pairs the model has no evidence for.

Bias is declared, not denied.

TWOSIDES and DrugBank skew toward US-marketed, English-labeled drugs. SIDER inherits FDA-label bias. PharmGuard outputs a Coverage Note on every report describing which databases were queried and what their known limitations are. The system never claims comprehensiveness — it claims fidelity to its sources.

The system is not a clinician.

Every output carries a disclaimer stating that PharmGuard is decision support, not medical judgment. The disclaimer is appended by the pipeline itself, not by the generator — so a prompt-injection attempt cannot remove it. The safety property is enforced by code, not by the model.

§ 9 · CONCLUSION

Disciplined scope as its own form of rigor.

PharmGuard promises one thing — drug-interaction detection grounded in structured evidence — and measures whether that one thing works. The Minimum Viable Logic is narrow by design: it does not attempt diagnosis, dosage optimization, or patient monitoring. In a landscape of overcommitted AI proposals, this discipline is a feature.

The architectural thesis is that certain cognitive labor — enumerating combinations, looking up records, classifying by thresholded statistics — is done more safely and more cheaply by deterministic code than by a probabilistic model. The LLM is reserved for the work where fluent natural-language synthesis is the actual product. This separation is what makes auditability possible: every deterministic step leaves a trace, and every generative claim leaves a citation.

The evaluation framework is self-consistent: ground truth is derived from whatever interaction table is loaded, so the same metric code produces meaningful numbers against the sample dataset, the full TWOSIDES release, or any intermediate subset. Failure modes — specifically, the silent-failure scenario — are defined in writing before success criteria. A system that scores 92% recall but 15% hallucination rate does not pass this project's bar, and the metric suite is engineered to ensure that.

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